

Sentinel AI

Task 1 and Task 2 Change Report

Prepared for: Ritwik Dhull

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Repository: Sentinel AI — Financial Fraud Detector

Executive summary

Task 1 established a safe, documented monorepo foundation. Task 2 converted it into a reproducible local shell with exactly four isolated services: FastAPI, React/TypeScript, PostgreSQL, and Redis. Transaction processing, schemas, streams, fraud scoring, datasets, ML, and AI remain deferred.

Task 1 — Repository foundation

- Added `AGENTS.md` with scope, safety, review, blocker, verification, and handoff rules.
- Added project and documentation READMEs.
- Collected the patent and task guides under `docs/planning`.
- Created boundaries for backend, frontend, packages, infrastructure, scripts, tests, data, and docs.
- Added placeholder-only environment documentation and comprehensive Git exclusions.
- Verified secrets, datasets, dependencies, caches, logs, models, and builds are excluded.

Outcome: an agent-ready repository without premature application, database, fraud, ML, or AI code.

Task 2 — Local platform shell

Backend

- Python 3.12 project locked with `uv`.
- FastAPI `/health` liveness and truthful `/ready` PostgreSQL/Redis checks.
- Short timeouts, no credential exposure, Ruff, strict mypy, pytest, five tests, and Dockerfile.

Frontend

- Node 22, React, TypeScript, Vite, and exact npm lockfile.
- Neutral operational page; no dashboard or product workflow.
- Prettier, ESLint, TypeScript, Vitest, production build, and Dockerfile.

Docker isolation

- One `sentinel-ai` Compose project with API, web, PostgreSQL, and Redis.
- Localhost-only ports, explicit tags, health checks, and dependency-aware startup.
- Dedicated network and persistent volumes; no UniHub resources referenced.

- Shutdown and cleanup preserve volumes.

Service	Address
Web	127.0.0.1:15173
API	127.0.0.1:18000
PostgreSQL	127.0.0.1:15432
Redis	127.0.0.1:16379

Commands

make bootstrap, make up, make ps, make logs, make verify, make down, and safe make clean.

Locked baseline

Area	Versions
Python	Python 3.12.13; uv 0.12.2; FastAPI 0.141.1; Uvicorn 0.52.1; asyncpg 0.31.0; redis 6.4.0
Frontend	Node 22.14.0; npm 10.9.2; React 19.2.8; Vite 8.2.1; TypeScript 5.9.3; Vitest 4.1.10
Containers	Python 3.12.13; uv 0.12.2; Node 22.14.0; PostgreSQL 16.4; Redis 7.2.5; local application tag :task2

Verification

- Locked bootstrap passed.
- Formatting, linting, strict typing, five backend tests, and one frontend test passed.
- Production build and npm audit passed with zero vulnerabilities.
- Compose validation/build/start and four health checks passed.
- Web, liveness, and readiness smoke tests passed.
- With Redis paused, liveness remained 200 and readiness correctly returned 503.
- Recovery, restart, volume persistence, logs, and clean shutdown passed.
- Final `make verify` passed.

Deferred work

- Transactions and ingestion APIs
- Business schemas and migrations
- Redis Streams and workers
- Authentication and product workflows
- Fraud scoring, alerts, datasets, ML, AI, and production deployment

Handoff

Task 1 and Task 2 are complete. The next planned task—the versioned transaction event contract and ingestion boundary—has not started.